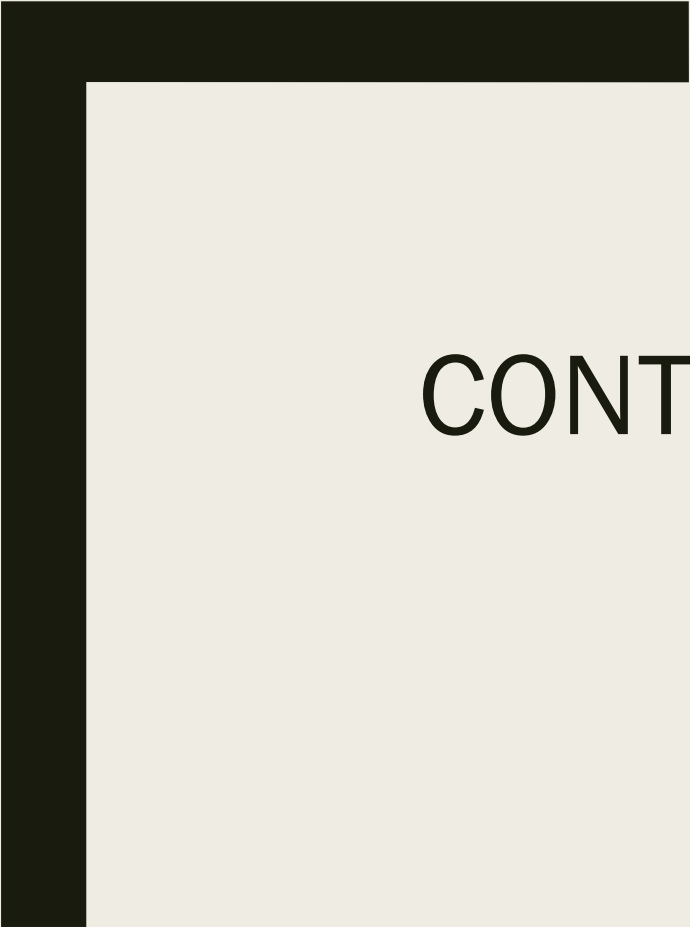
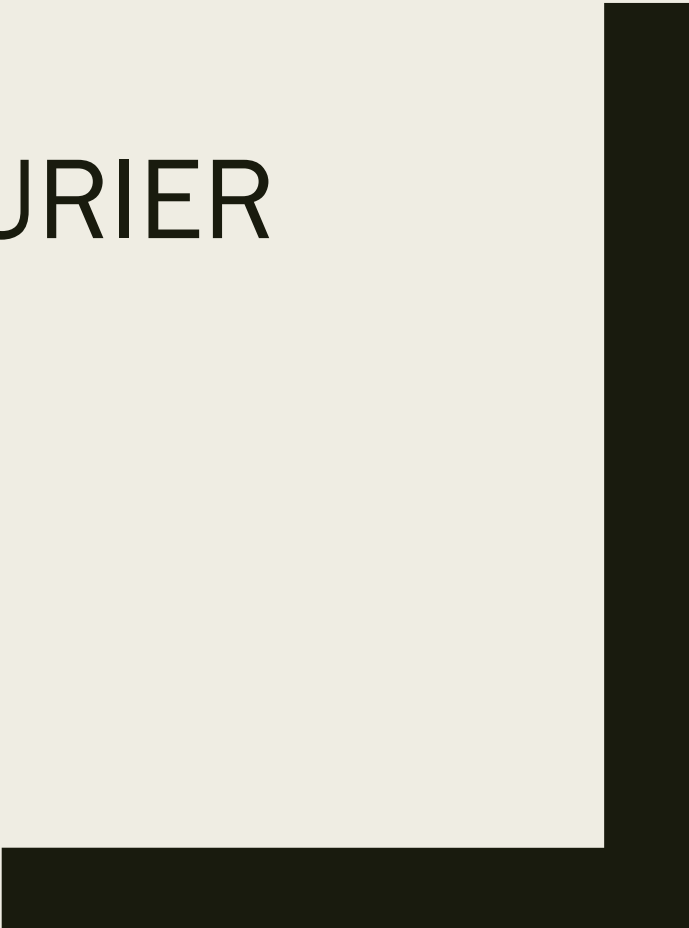




LEC 9.1: CONTINUOUS TIME FOURIER SERIES (CTFS)

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Transformation Methods for CT signals

- A real time naturally available signal is in the form of *time domain*
- However, analysis of a signal is far more convenient in *frequency domain*.
- A signal in its original form can not be defined in terms of frequency. Hence, by transformation methods, a time domain signal can be represented with respect to frequency components.
- The three important transformation methods available in practice for a continuous time domain signals are as follows
 - Fourier Series
 - Fourier Transform
 - Laplace Transform

Fourier Series

- The representation of signals over a certain interval of time in terms of the linear combination of the **orthogonal functions** is called Fourier Series.
- Fourier Series expansion is used for **periodic signals** to expand them in terms of their **harmonics** which are sinusoidal and orthogonal to each other.
- Two important classes of Fourier series methods are primarily available. They are as follows;
 - Exponential Fourier Series
 - If orthogonal functions are exponential functions (written in form of $e^{j\omega t}$)
 - Trigonometric Fourier Series
 - If orthogonal functions are trigonometric functions (written in the form of $\sin nx$, $\cos mx$)

- **Orthogonal Functions:** Two functions (signals) f and g are said to be orthogonal when their dot product is zero.

- $\int_{-\infty}^{\infty} f(t)g(t) dt = 0$

- **Periodic Signal:** Signal which repeats itself after a fixed interval of time is called periodic signal.

- $x(t) = x(t + T)$

- **Harmonics:** A **harmonic** is a signal or wave whose frequency is an integer multiple of the fundamental frequency.

- Suppose f_o is fundamental frequency of any signal. Thus, $2f_o, 4f_o, 6f_o$ etc. are even harmonics of the signal while $3f_o, 5f_o, 7f_o$ etc. are odd harmonics of the signal

Derivative of Complex Exponential Fourier Series

- A complex exponential signal can be written as,
- $x(t) = ae^{j\omega_0 t} = ae^{j2\pi f_0 t}$
- If we take the signal at various intervals of the fundamental period or at various harmonics, then:
 - the constant signal or dc signal is given by $x_0(t) = a_0 e^{j2\pi(0)t} = a_0$
 - The first harmonic of the signal is given by $x_1(t) = a_1 e^{j2\pi(1f_0)t} = a_1 e^{j2\pi f_0 t} = a_1 e^{j\omega_0 t}$
 - The second harmonic of the signal is given by $x_2(t) = a_2 e^{j2\pi(2f_0)t} = a_2 e^{j(2\omega_0)t}$
 - The third harmonic of the signal is given by $x_3(t) = a_3 e^{j2\pi(3f_0)t} = a_3 e^{j(3\omega_0)t}$
 - Similarly for k th harmonic, $x_k(t) = a_k e^{j2\pi(kf_0)t} = a_k e^{j(k\omega_0)t}$

- Thus, the linear combination of these complex exponential can be written as;
- $x(t) = x_o(t) + x_1(t) + x_2(t), \dots x_k(t)$
- $x(t) = a_o + a_1 e^{j\omega_o t} + a_2 e^{j(2\omega_o)t} + \dots a_k e^{j(k\omega_o)t}$

$$\blacksquare \quad x(t) = \sum_{k=-\infty}^{\infty} a_k e^{j(k\omega_o)t} \quad (i)$$

- Eq. (i) is also known as **synthesis equations**. Here k is an integer value whereas a_k is the weight of each harmonic and is known as **Fourier series coefficients** or spectral coefficients.
- Determination of Fourier series is nothing but determination of **Fourier series coefficients**.

How to find Fourier Series (FS) Coefficients

- $a_0 = \frac{1}{T_0} \int_0^{T_0} x(t) dt \quad (ii)$

- $a_k = \frac{1}{T_0} \int_0^{T_0} x(t) e^{-j(k\omega_0)t} dt \quad (iii)$

- Eq. (ii) and (iii) are known as Analysis Equation

Exp 1: (Exp 3.3): Write FS expansion of the signal
 $x(t) = \sin \omega_o t$

■ **Solution:** To find the FS expansion, we need to determine the FS coefficients i.e. a_o and a_k

■ **Method 1:** Use analysis equation to find the coefficients

■ $a_o = \frac{1}{T_o} \int_0^{T_o} x(t) dt$ and $a_k = \frac{1}{T_o} \int_0^{T_o} x(t) e^{-j(k\omega_o)t} dt$

■ **Method 2:** It is a simple case. We can expand the sinusoidal signal into exponentials.

■ $x(t) = \sin \omega_o t = \frac{e^{j\omega_o t} - e^{-j\omega_o t}}{2j}$

■ $x(t) = \frac{e^{j\omega_o t}}{2j} + \frac{e^{-j\omega_o t}}{-2j}$

- $x(t) = \frac{1}{2j} e^{j\omega_o t} + \frac{1}{-2j} e^{-j\omega_o t}$
- By inspection, we can find that when
- $k = 1, \quad e^{jk\omega_o t} \rightarrow e^{j\omega_o t}$
- $k = -1, \quad e^{jk\omega_o t} \rightarrow e^{-j\omega_o t}$
- Therefore,
- $a_1 = \frac{1}{2j}$ and $a_{-1} = -\frac{1}{2j}$
- $a_k = 0$ when $k \neq +1$ and -1

Exp 2: Write FS expansion of the signal

$$x(t) = \sin\left(2t + \frac{\pi}{3}\right)$$

- Solution: It is a simple case. We can expand the sinusoidal signal into exponentials.

- $$x(t) = \sin\left(2t + \frac{\pi}{3}\right) = \frac{e^{j\left(2t + \frac{\pi}{3}\right)} - e^{-j\left(2t + \frac{\pi}{3}\right)}}{2j}$$

- $$x(t) = \left(\frac{1}{2j}\right) e^{j\left(2t + \frac{\pi}{3}\right)} + \left(-\frac{1}{2j}\right) e^{-j\left(2t + \frac{\pi}{3}\right)}$$

- $$x(t) = \left(\frac{1}{2j} e^{j\frac{\pi}{3}}\right) e^{j(2t)} + \left(-\frac{1}{2j} e^{-j\frac{\pi}{3}}\right) e^{-j(2t)}$$

- By inspection, we can find that when

- $k = 2, \quad e^{jk\omega_0 t} \rightarrow e^{j(2t)} \quad k(1)t = 2t$

- $k = -2,$

- Therefore,
- $a_2 = \left(\frac{1}{2j} e^{j\frac{\pi}{3}}\right)$ and $a_{-2} = \left(-\frac{1}{2j} e^{j\frac{\pi}{3}}\right)$
- $a_k = 0$ when $k \neq +2$ and -2

Exp 3: Write FS expansion of the signal

$$x(t) = \cos 3t + \sin 6t$$

- $x(t) = \cos 3t + \sin 6t = \frac{e^{j(3t)} + e^{-j(3t)}}{2} + \frac{e^{j(6t)} - e^{-j(6t)}}{2j}$
- $x(t) = \frac{1}{2}e^{j(3t)} + \frac{1}{2}e^{-j(3t)} + \frac{1}{2j}e^{j(6t)} + \frac{1}{-2j}e^{-j(6t)}$
- By inspection, we can find that when $k(1)t = 6t$
- $k = 3, k = -3, k = 6, k = -6,$
- Therefore,
- $a_3 = \left(\frac{1}{2}\right), a_{-3} = \left(\frac{1}{2}\right), a_6 = \left(\frac{1}{2j}\right), a_{-6} = \left(\frac{1}{-2j}\right)$
- $a_k = 0$ when $k \neq |3|$ and $|6|$

Exp 4: (Exp 3.4): Write FS expansion of the given signal and also draw their magnitude and phase spectrum

$$x(t) = 1 + \sin \omega_o t + 2 \cos \omega_o t + \cos \left(2\omega_o t + \frac{\pi}{4} \right)$$

- $x(t) = 1 + \left(\frac{1}{2j} \right) [e^{j(\omega_o t)} - e^{-j(\omega_o t)}] + 2 \left(\frac{1}{2} \right) [e^{j(\omega_o t)} + e^{-j(\omega_o t)}] + \left(\frac{1}{2} \right) [e^{j(2\omega_o t + \frac{\pi}{4})} + e^{-j(2\omega_o t + \frac{\pi}{4})}]$
- $x(t) = 1 + \left(\frac{1}{2j} \right) [e^{j(\omega_o t)} - e^{-j(\omega_o t)}] + (1) [e^{j(\omega_o t)} + e^{-j(\omega_o t)}] + \left(\frac{1}{2} e^{j(\frac{\pi}{4})} \right) e^{j(2\omega_o t)} + \left(\frac{1}{2} e^{-j(\frac{\pi}{4})} \right) e^{-j(2\omega_o t)}$
- Collect terms with similar exponential powers
- $x(t) = 1 + \left(1 + \frac{1}{2j} \right) e^{j(\omega_o t)} + \left(1 - \frac{1}{2j} \right) e^{-j(\omega_o t)} + \left(\frac{1}{2} e^{j(\frac{\pi}{4})} \right) e^{j(2\omega_o t)} + \left(\frac{1}{2} e^{-j(\frac{\pi}{4})} \right) e^{-j(2\omega_o t)}$
- $a_o = 1, a_1 = \left(1 + \frac{1}{2j} \right), a_{-1} = \left(1 - \frac{1}{2j} \right), a_2 = \left(\frac{1}{2} e^{j(\frac{\pi}{4})} \right), a_{-2} = \left(\frac{1}{2} e^{-j(\frac{\pi}{4})} \right)$

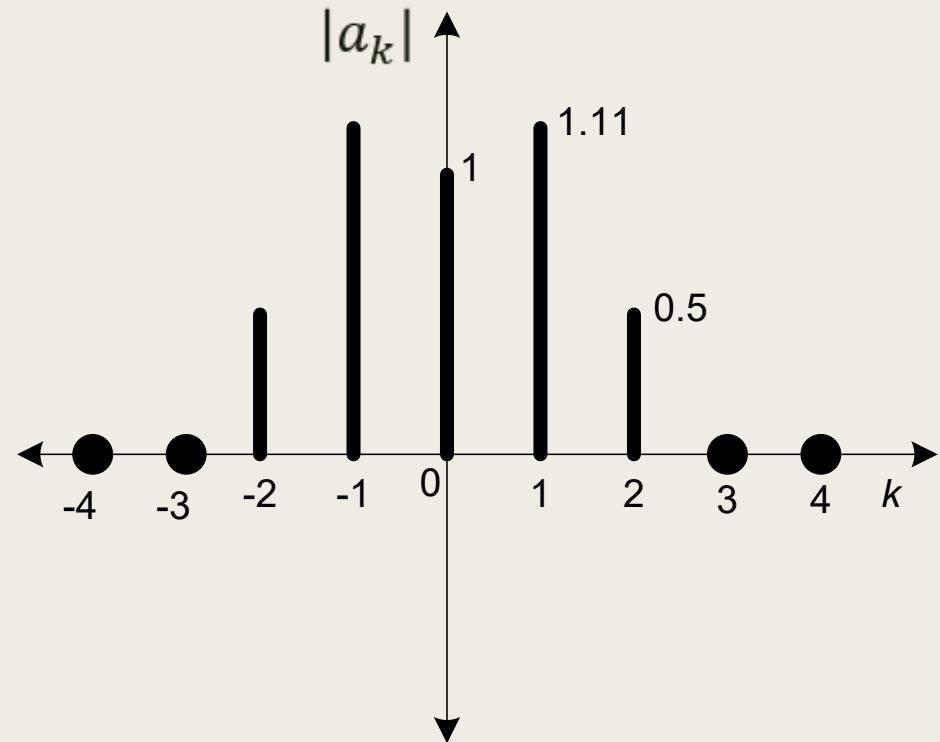
- $a_0 = 1$
- $a_{+1} = \left(1 + \frac{1}{2j}\right) = 1 - \frac{1}{2}j$
- $a_{-1} = \left(1 - \frac{1}{2j}\right) = 1 + \frac{1}{2}j$
- $a_2 = \left(\frac{1}{2}e^{j\left(\frac{\pi}{4}\right)}\right) = \frac{1}{2}(\cos\frac{\pi}{4} + jsin\frac{\pi}{4}) = \frac{1}{2}(\cos 45^\circ + jsin45^\circ) = \frac{1}{2}\left(\frac{\sqrt{2}}{2} + j\frac{\sqrt{2}}{2}\right) = \frac{\sqrt{2}}{4}(1 + j)$
- $a_2 = \frac{\sqrt{2}}{4}(1 + j)$
- $a_{-2} = \left(\frac{1}{2}e^{-j\left(\frac{\pi}{4}\right)}\right) = \frac{1}{2}(\cos\frac{\pi}{4} - jsin\frac{\pi}{4}) = \frac{1}{2}(\cos 45^\circ - jsin45^\circ) = \frac{1}{2}\left(\frac{\sqrt{2}}{2} - j\frac{\sqrt{2}}{2}\right) = \frac{\sqrt{2}}{4}(1 - j)$
- $a_{-2} = \frac{\sqrt{2}}{4}(1 - j)$
- $a_k = 0 \text{ for } |k| > 2$

Magnitude and Phase Spectrum

- Magnitude Spectrum $|a_k| = \sqrt{Re^2 + Img^2}$
- Phase Spectrum $\angle a_k = \tan^{-1} \left(\frac{Img}{Re} \right)$
- **Magnitude Spectrum** Now for different values of k
- $|a_0| = \sqrt{1^2 + 0^2} = 1$
- $|a_1| = \sqrt{1^2 + \left(-\frac{1}{2}\right)^2} = \sqrt{1 + \frac{1}{4}} = \sqrt{5/4} = 1.11$
- $|a_{-1}| = \sqrt{1^2 + \left(\frac{1}{2}\right)^2} = \sqrt{1 + \frac{1}{4}} = \sqrt{5/4} = 1.11$

- $|a_2| = \sqrt{\left(\frac{\sqrt{2}}{4}\right)^2 + \left(\frac{\sqrt{2}}{4}\right)^2} = \sqrt{\frac{2}{16} + \frac{2}{16}} = \sqrt{4/16} = \frac{2}{4} = 0.5$

- $|a_{-2}| = \sqrt{\left(\frac{\sqrt{2}}{4}\right)^2 + \left(-\frac{\sqrt{2}}{4}\right)^2} = \sqrt{\frac{2}{16} + \frac{2}{16}} = \sqrt{4/16} = \frac{2}{4} = 0.5$



■ Phase Spectrum

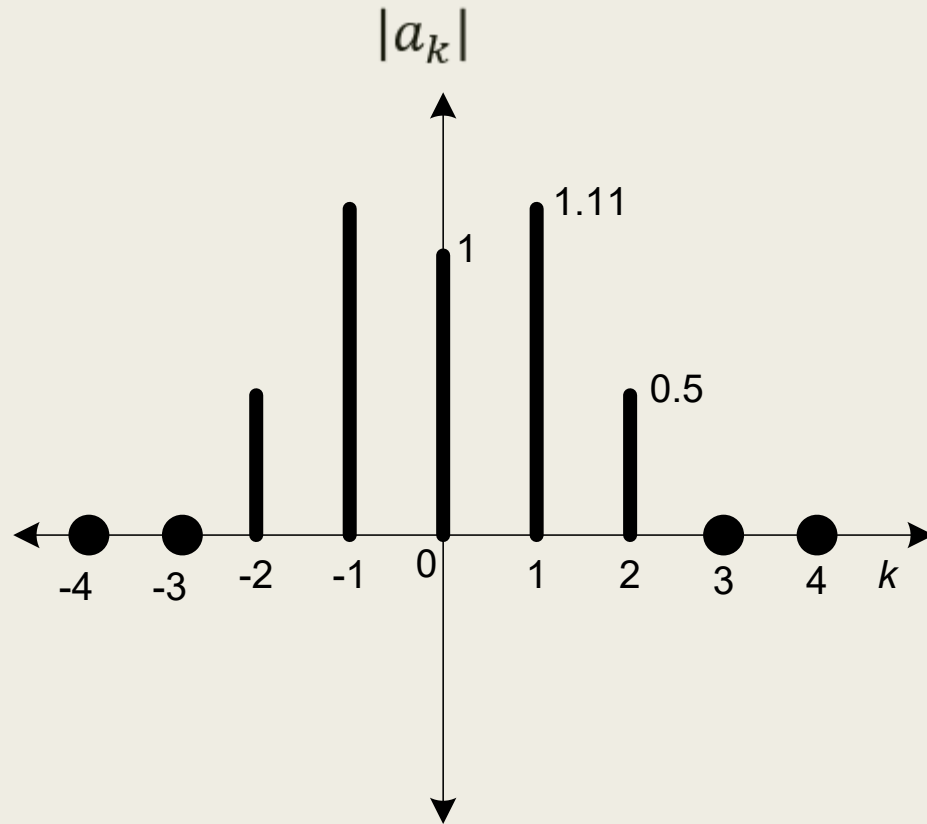
$$\blacksquare a_o = \tan^{-1} \left(\frac{\text{Im}g}{\text{Re}} \right) = \tan^{-1} \left(\frac{0}{1} \right) = \tan^{-1}(0) = 0$$

$$\blacksquare a_1 = \tan^{-1} \left(\frac{\text{Im}g}{\text{Re}} \right) = \tan^{-1} \left(\frac{-1/2}{1} \right) = \tan^{-1}(-0.5) = -0.46$$

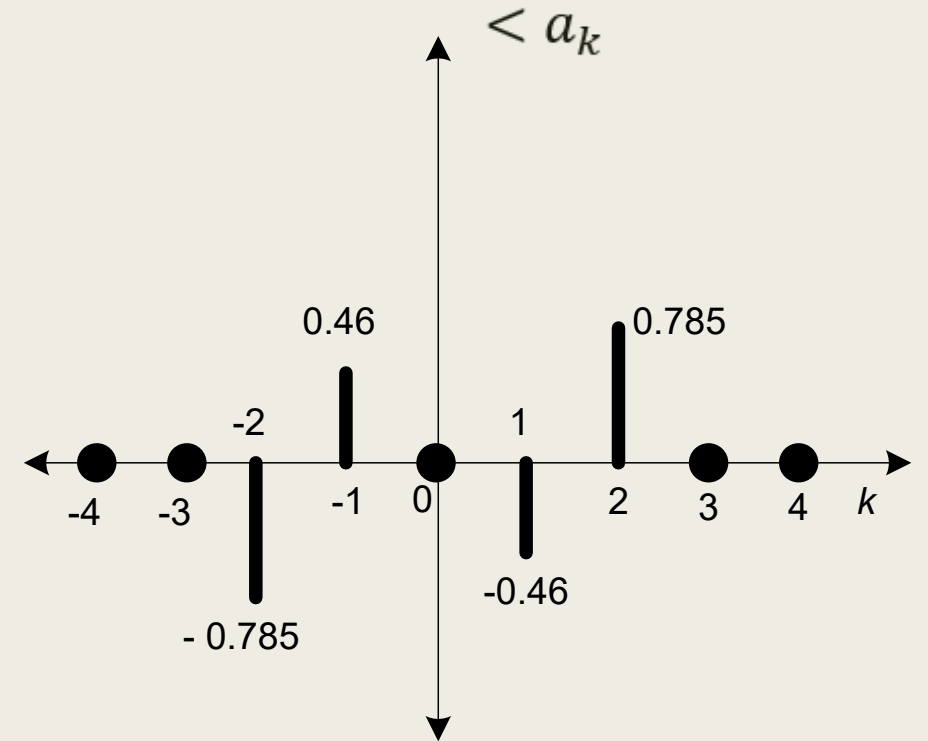
$$\blacksquare a_{-1} = \tan^{-1} \left(\frac{\text{Im}g}{\text{Re}} \right) = \tan^{-1} \left(\frac{1/2}{1} \right) = \tan^{-1}(0.5) = 0.46$$

$$\blacksquare a_2 = \tan^{-1} \left(\frac{\text{Im}g}{\text{Re}} \right) = \tan^{-1} \left(\frac{\frac{\sqrt{2}}{4}}{\frac{\sqrt{2}}{4}} \right) = \tan^{-1}(1) = \frac{\pi}{4} = 0.785$$

$$\blacksquare a_{-2} = \tan^{-1} \left(\frac{\text{Im}g}{\text{Re}} \right) = \tan^{-1} \left(\frac{-\frac{\sqrt{2}}{4}}{\frac{\sqrt{2}}{4}} \right) = \tan^{-1}(-1) = -\frac{\pi}{4} = -0.785$$



Magnitude Spectrum follows even symmetry



Phase Spectrum follows odd symmetry

Thank You !!!